

实验二

一、实验目的

- 1、掌握 Shell 脚本的基本语法与结构，包括变量赋值、条件判断、循环控制等。
- 2、理解 Shell 函数的概念与使用，学会定义并调用函数。
- 3、熟悉 Linux 环境下 Shell 脚本的编辑、调试与执行流程。
- 4、通过编写实用脚本解决实际问题，提升脚本编程的实践能力。

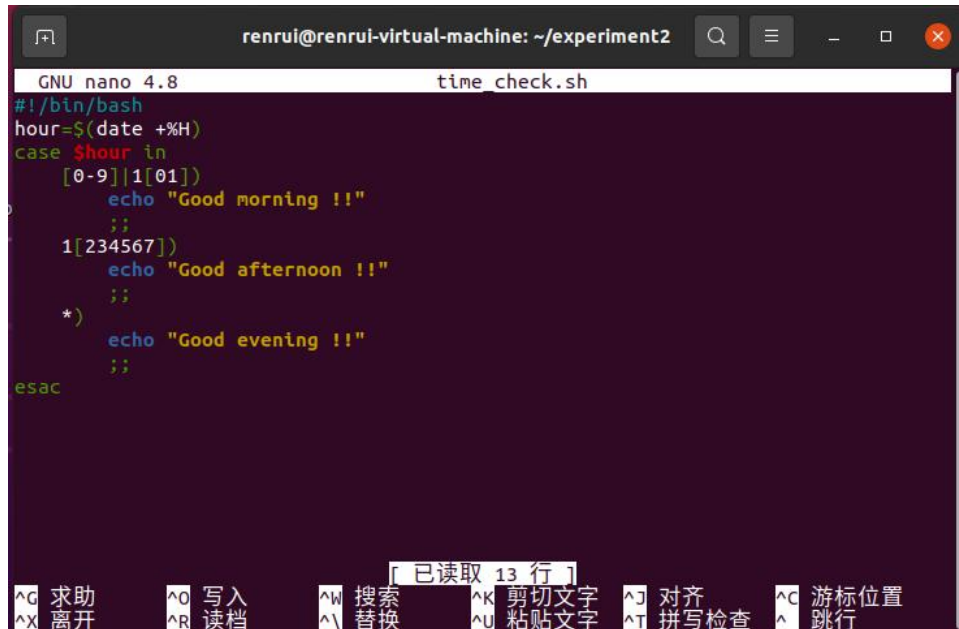
二、实验内容

- 1、获取系统时间，并判断是上午、下午还是晚上

创建一个专门用于本次实验的目录，并进入该目录，创建脚本文件 time_check.sh

```
renrui@renrui-virtual-machine:~/桌面$ mkdir ~/experiment2
renrui@renrui-virtual-machine:~/桌面$ cd ~/experiment2
renrui@renrui-virtual-machine:~/experiment2$ nano time_check.sh
```

输入代码



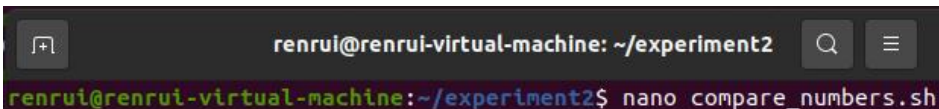
```
GNU nano 4.8 time_check.sh
#!/bin/bash
hour=$(date +%H)
case $hour in
    [0-9]|1[01])
        echo "Good morning !!"
        ;;
    1[234567])
        echo "Good afternoon !!"
        ;;
    *)
        echo "Good evening !!"
        ;;
esac
```

保存并退出后，赋予执行权限并运行，分别在不同时间段测试，观察实验结果是否符合预期

```
renrui@renrui-virtual-machine:~/experiment2$ nano time_check.sh
renrui@renrui-virtual-machine:~/experiment2$ chmod +x time_check.sh
renrui@renrui-virtual-machine:~/experiment2$ ./time_check.sh
Good evening !!
```

- 2、比较两个数字的大小

创建脚本文件 compare_numbers.sh



```
renrui@renrui-virtual-machine: ~/experiment2
renrui@renrui-virtual-machine:~/experiment2$ nano compare_numbers.sh
```

输入代码

```
renrui@renrui-virtual-machine: ~/experiment2
GNU nano 4.8 compare_numbers.sh
#!/bin/bash
echo "Enter the first integer:"
read first
echo "Enter the second integer:"
read second
if [ "$first" -gt "$second" ]; then
    echo "$first is greater than $second"
elif [ "$first" -lt "$second" ]; then
    echo "$first is less than $second"
else
    echo "$first is equal to $second"
fi
```

保存并退出，赋予执行权限并运行

分别测试了 2 与 10、5 与 5、15 与 13 的大小，结果显示正确

```
renrui@renrui-virtual-machine:~/experiment2$ chmod +x compare_numbers.sh
renrui@renrui-virtual-machine:~/experiment2$ ./compare_numbers.sh
Enter the first integer:
2
Enter the second integer:
10
2 is less than 10
renrui@renrui-virtual-machine:~/experiment2$ ./compare_numbers.sh
Enter the first integer:
5
Enter the second integer:
5
5 is equal to 5
renrui@renrui-virtual-machine:~/experiment2$ ./compare_numbers.sh
Enter the first integer:
15
Enter the second integer:
13
15 is greater than 13
renrui@renrui-virtual-machine:~/experiment2$
```

3、找出列表中最小值

创建脚本文件 find_min.sh

```
renrui@renrui-virtual-machine:~/experiment2$ nano find_min.sh
renrui@renrui-virtual-machine:~/experiment2$
```

输入代码

```
renrui@renrui-virtual-machine: ~/experiment2
GNU nano 4.8 find_min.sh
#!/bin/bash
smallest=10000
for i in 8 2 18 0 -3 87
do
    if [ $i -lt $smallest ]; then
        smallest=$i
    fi
done
echo $smallest
```

保存并退出，赋予执行权限并运行，结果显示为-3，正确

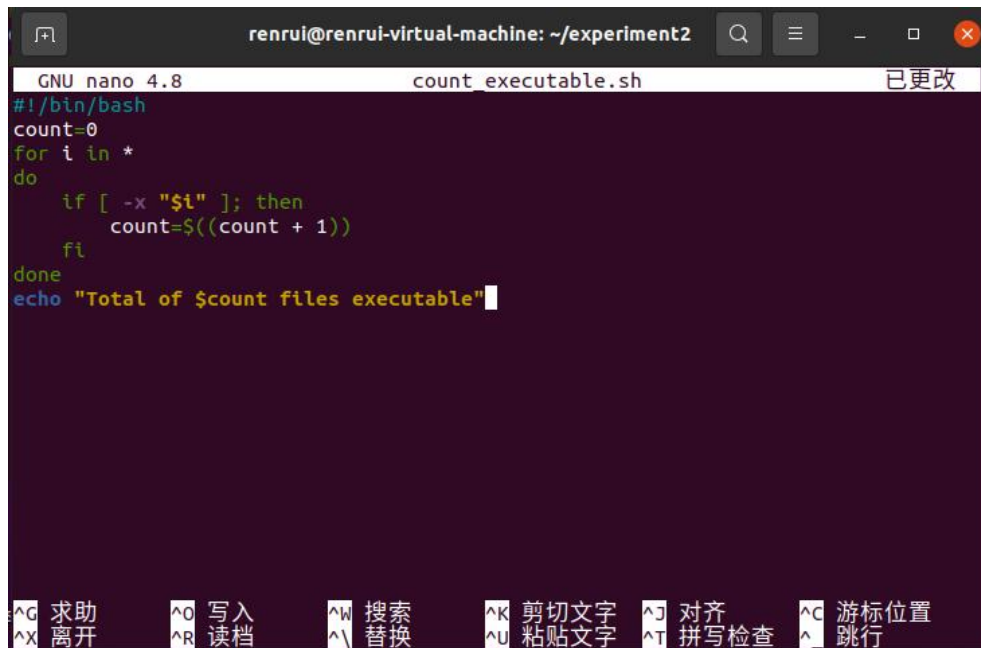
```
renrui@renrui-virtual-machine:~/experiment2$ chmod +x find_min.sh
renrui@renrui-virtual-machine:~/experiment2$ ./find_min.sh
3
renrui@renrui-virtual-machine:~/experiment2$
```

4、计算当前目录中可执行文件数量

创建脚本文件 count_executable.sh

```
renrui@renrui-virtual-machine:~/experiment2$ nano count_executable.sh
renrui@renrui-virtual-machine:~/experiment2$
```

输入代码



```
renrui@renrui-virtual-machine: ~/experiment2
GNU nano 4.8 count_executable.sh 已更改
#!/bin/bash
count=0
for i in *
do
    if [ -x "$i" ]; then
        count=$((count + 1))
    fi
done
echo "Total of $count files executable"
```

底部状态栏显示快捷键：^G 求助, ^X 离开, ^O 写入, ^R 读档, ^W 搜索, ^\ 替换, ^K 剪切文字, ^U 粘贴文字, ^J 对齐, ^T 拼写检查, ^C 光标位置, ^_ 跳行

保存并退出，赋予执行权限并运行，结果显示有 4 个，输入 ls 进行验证，结果正确

```
renrui@renrui-virtual-machine:~/experiment2$ chmod +x count_executable.sh
renrui@renrui-virtual-machine:~/experiment2$ ./count_executable.sh
Total of 4 files executable
renrui@renrui-virtual-machine:~/experiment2$ ls
compare_numbers.sh count_executable.sh find_min.sh time_check.sh
renrui@renrui-virtual-machine:~/experiment2$
```

5、判断一个数是否为素数

创建脚本文件 prime_check.sh

```
renrui@renrui-virtual-machine:~/experiment2$ nano prime_check.sh
renrui@renrui-virtual-machine:~/experiment2$
```

输入代码

```
renrui@renrui-virtual-machine: ~/experiment2
GNU nano 4.8 prime_check.sh 已更改
#!/bin/bash
prime() {
    flag=1
    j=2
    while [ $j -le $(( $1 / 2 )) ]
    do
        if [ $(( $1 % $j )) -eq 0 ]; then
            flag=0
            break
        fi
        j=$((j + 1))
    done
    if [ $flag -eq 1 ]; then
        return 1
    else
        return 0
    fi
}

if [ $# -eq 0 ]; then
```

```
renrui@renrui-virtual-machine: ~/experiment2
GNU nano 4.8 prime_check.sh
done
if [ $flag -eq 1 ]; then
    return 1
else
    return 0
fi
}

if [ $# -eq 0 ]; then
    echo "Usage: ./prime_check.sh <number>"
    exit 1
fi

prime $1
if [ $? -eq 1 ]; then
    echo "$1 is a prime!"
else
    echo "$1 is not a prime!"
fi
```

保存并退出，赋予执行权限并运行，测试数为 20 和 17，结果正确

```
renrui@renrui-virtual-machine:~/experiment2$ chmod +x prime_check.sh
renrui@renrui-virtual-machine:~/experiment2$ ./prime_check.sh 17
17 is a prime!
renrui@renrui-virtual-machine:~/experiment2$ ./prime_check.sh 20
20 is not a prime!
renrui@renrui-virtual-machine:~/experiment2$
```

三、实验小结

通过本次实验，我掌握了 Shell 脚本的基本语法与控制结构，能够编写实现时间判断、数值比较、最小值查找、文件统计和素数判断等功能的脚本。实验中遇到的主要问题包括时间匹配语法错误、变量名大小写不一致以及函数返回值逻辑混淆，通过查阅资料和调试逐一解决。本次实践不仅加深了我对 Shell 编程的理解，也提升了在 Linux 环境下编写、调试脚本的实际操作能力，为后续学习系统管理和自动化脚本开发奠定了基础。

